Hints: Linear algebra – 10. Matrix norms and localisation of the spectrum

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Lower bound: let $Av = \lambda v$ and $X = (v, v, \dots, v)$. Then $A^k X = \lambda^k X$, so $|\lambda|^k \|X\| \leqslant \|A^k\|\,\|X\|$. This gives $\rho(A) \leqslant \|A^k\|^{1/k}$ for all k; $k = 1$ is the claim $\rho(A) \leqslant \|A\|$.

Hint 2. Upper bound: for $\varepsilon > 0$ the matrix $B = A/(\rho(A) + \varepsilon)$ has spectral radius < 1. The Jordan form shows that $B^k \to O$ (entries are polynomial in k times μ^k with $|\mu| < 1$). So $\|A^k\| \leqslant C(\rho + \varepsilon)^k$ in any norm, since all norms are equivalent.

Hint 3. Norm with $\|A\| \leqslant \rho + \varepsilon$: conjugate the Jordan form by $\operatorname{diag}(1, \delta, \delta^2, \dots)$ to make the off-diagonal entries δ. Then use the operator norm induced by $\|x\| := \|S^{-1}x\|_\infty$ for the corresponding change of basis S. Alternatively, use the vector norm $\|x\| := \sum_k \|A^k x\|/(\rho + \varepsilon)^k$.

Problem 2

Hint 1. Triangularise A and B simultaneously over $\mathbb{C}$. The eigenvalues of $A+B+E$ are $\alpha_i+\beta_i+1$ with $\alpha_i^{2002} = 1$ and $\beta_i^{2003} = 1$.

Hint 2. Two unit complex numbers with $\alpha + \beta = -1$ must be $e^{\pm 2\pi i/3}$, so $\alpha^3 = 1$. But the order of α divides $\gcd(3, 2002) = 1$.

Problem 3

Hint 1. Put $a_k = \|A^k - A^{k+1}\|$ for $k \geqslant 0$, so that $a_k \leqslant \frac{c}{k+1}$ with $c = \frac{1}{2002}$. Show that $a_{2k} - a_{2k+1} \leqslant a_k^2$ and $a_{2k+1} - a_{2k+2} \leqslant a_k a_{k+1}$. Hint: $A^{2k}(E - A)^2 = (A^k(E - A))^2$.

Hint 2. Since $a_N \to 0$, $a_m = \sum_{j \geqslant m}(a_j - a_{j+1}) \leqslant \sum_{k \geqslant \lfloor m/2 \rfloor}(a_k^2 + a_k a_{k+1})$.

Hint 3. If $a_k \leqslant \frac{c}{k+1}$ for all k, the previous estimate gives $a_m \leqslant \frac{12c^2}{m+1}$ for all m. With $c < 1/12$, iterating $c \mapsto 12c^2$ drives the constant to 0. So all $a_k = 0$; in particular $A = E$, and the claim is immediate.

Problem 4

Hint 1. Let $r = \|A\|$. If $r \leqslant n/\ln 2$, the trivial bound $\|A^n\| \leqslant r^n = r \cdot r^{n-1}$ suffices.

Hint 2. Otherwise use Cayley–Hamilton: $A^n = -\sum_{k<n} c_k A^k$, where c_k are the coefficients of $\prod(x - \lambda_i)$. Since $|\lambda_i| \leqslant 1$, $|c_k| \leqslant \binom{n}{k}$. Hence $\|A^n\| \leqslant (1 + r)^n - r^n = r^n\big((1 + \frac{1}{r})^n - 1\big)$.

Hint 3. With $x = n/r \leqslant \ln 2$: $(1 + \frac{1}{r})^n - 1 \leqslant e^x - 1 \leqslant x/\ln 2$, because $(e^x - 1)/x$ is increasing and equals $1/\ln 2$ at $x = \ln 2$.

Problem 5

Hint 1. Compute $\operatorname{tr}(A^2) = \sum_{i,j} a_{ij} a_{ji}$. The hypothesis with $i = j$ forces $a_{ii} = 0$, so $\operatorname{tr} A^2 \leqslant 0$.

Hint 2. If all eigenvalues were real, $\sum \lambda_i^2 = \operatorname{tr} A^2 \leqslant 0$ would force all $\lambda_i = 0$. Then A would be nilpotent and $A^n = O$. Non-real eigenvalues of a real matrix come in conjugate pairs.